

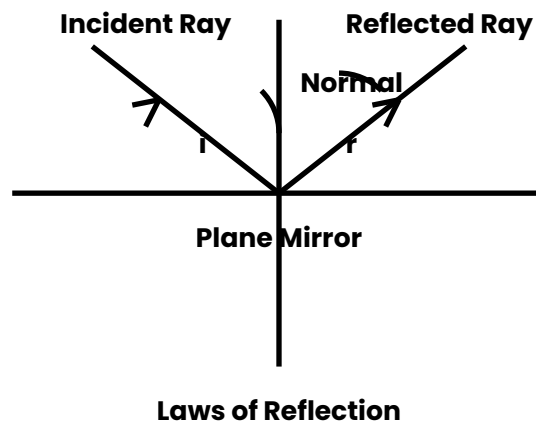
REFLECTION

Reflection of Light

1. Reflection of Light

Reflection is the phenomenon in which light returns into the same medium after striking a reflecting surface. The incident ray falls on the surface, while the reflected ray returns from the surface.

- The incident ray is the ray falling on the reflecting surface.
- The reflected ray is the ray that returns from the reflecting surface.
- The normal is an imaginary line perpendicular to the reflecting surface at the point of incidence.

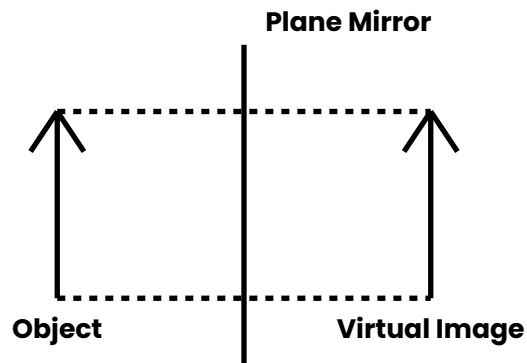


2. Reflection from Plane Mirror

A plane mirror has a flat reflecting surface and forms an image behind the mirror.

- The image formed by a plane mirror is virtual and erect.
- The image is of the same size as the object.
- The image is laterally inverted.

- The image is formed at the same distance behind the mirror as the object is in front of it.

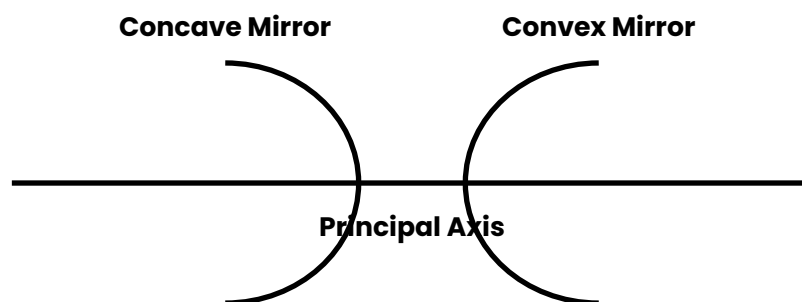


Plane Mirror Image Formation

3. Reflection of Light by Spherical Mirrors

A spherical mirror is a part of a hollow sphere whose one surface is polished. Spherical mirrors are of two types: concave mirror and convex mirror.

- Concave mirror has its reflecting surface on the inner side.
- Convex mirror has its reflecting surface on the outer side.
- Important terms are pole (P), centre of curvature (C), radius of curvature (R), principal focus (F) and principal axis.

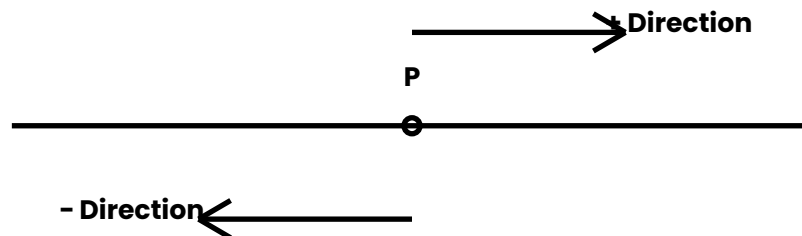


Spherical Mirrors

4. Sign Convention

The Cartesian sign convention is used for spherical mirrors. All distances are measured from the pole of the mirror.

- Distances measured in the direction of incident light are taken as positive.
- Distances measured opposite to the direction of incident light are taken as negative.
- Heights measured above the principal axis are positive.
- Heights measured below the principal axis are negative.



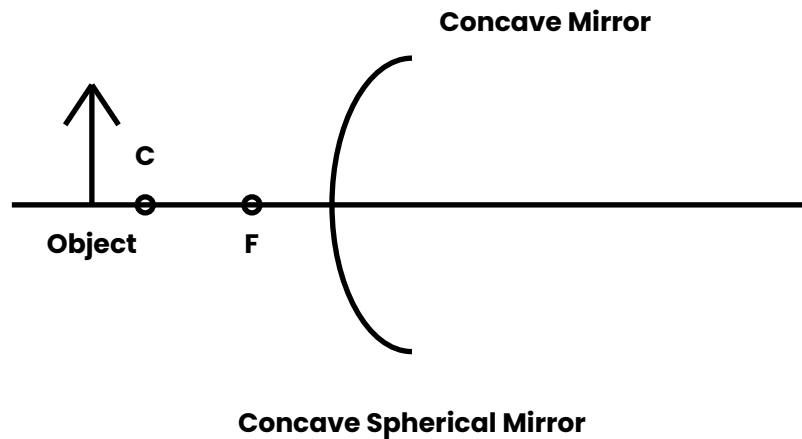
Cartesian Sign Convention

5. Image Formation by Spherical Mirrors

Concave Mirror

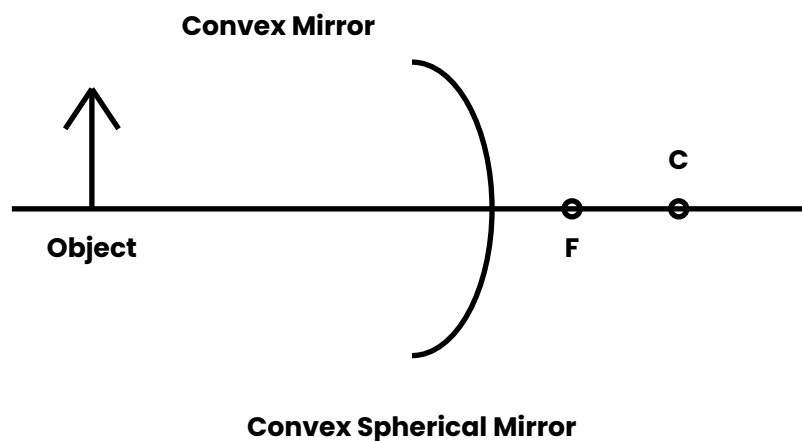
- Object beyond C: image forms between C and F and is real, inverted and diminished.
- Object at C: image forms at C and is real, inverted and of the same size.
- Object between C and F: image forms beyond C and is real, inverted and magnified.
- Object at F: image is formed at infinity.

- Object between F and P: image is virtual, erect and magnified.



Convex Mirror

- A convex mirror always forms a virtual, erect and diminished image.



6. Mirror Formula

The mirror formula gives the relation between object distance, image distance and focal length of a spherical mirror.

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$R = 2f$$

7. Magnification

Magnification is the ratio of the height of the image to the height of the object.

$$m = \frac{h_i}{h_o}$$

$$m = -v / u$$

- Negative magnification indicates an inverted image.
- Positive magnification indicates an erect image.

8. Uses of Spherical Mirrors

- Concave mirrors are used as shaving mirrors and by dentists because they can produce magnified images.
- Concave mirrors are used as reflectors in headlights, searchlights and torches.
- Convex mirrors are used as rear-view mirrors because they provide a wider field of view.

9. Important Formulae

$$1/f = 1/u + 1/v$$

$$m = \frac{h_i}{h_o}$$

$$m = -v / u$$

$$R = 2f$$

$$f = R / 2$$

10. Quick Revision Points

- Concave mirror can form real or virtual images depending on the object position.
- Convex mirror always forms a virtual, erect and diminished image.
- The principal focus F lies midway between the pole P and centre of curvature C .
- The principal axis passes through the pole and centre of curvature.
- Always apply the Cartesian sign convention carefully in numerical problems.

11. MHT-CET Important Points / Numerical Focus

- Mirror formula based numericals are important.
- Magnification based numericals are important.
- Questions involving sign convention are calculation-oriented.
- Image position, nature and size should be identified from the object position.
- Remember the relation $R = 2f$ for spherical mirrors.